

## No. of Functional Features Included in the Solution

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 03 July 2026                    |
| <b>Team ID</b>       |                                 |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 5 Marks                         |

### Functional Features Overview:

This document lists all the functional features that have been included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution. The number and quality of functional features reflect the depth and completeness of the project.

| S. No | Feature Name                                         | Feature Description                                                                                                                                                                                                                   | Module / Component                                 | Status (Done/In Progress /Pending) | Marks Contribution |
|-------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------|--------------------|
| 1     | <b>Dataset Loading &amp; Merging</b>                 | Load application_record.csv (438,557 rows) and credit_record.csv (1M+ rows) from Kaggle, merge on applicant ID to create a unified 36,457-record dataset with binary risk labels derived from payment STATUS codes.                   | 01_data_exploration .ipynb Epic 1: Data Collection | Done                               | Core Feature       |
| 2     | <b>Exploratory Data Analysis (EDA)</b>               | Generate 6 visualizations: class imbalance chart, income distribution, age distribution, risk rate by gender/education/income type, employment analysis, and correlation heatmap to understand data patterns before modeling.         | 02_eda.ipynb Epic 2: Visualization & Analysis      | Done                               | Core Feature       |
| 3     | <b>Data Pre-processing &amp; Feature Engineering</b> | Handle 134,203 missing OCCUPATION_TYPE values, fix DAYS_EMPLOYED placeholder (365243), engineer AGE_YEARS, YEARS_EMPLOYED, IS_UNEMPLOYED features, apply one-hot encoding to 5 categorical columns, expanding from 19 to 47 features. | 03_preprocessing .ipynb Epic 3: Pre-processing     | Done                               | Core Feature       |

| S. No | Feature Name                                       | Feature Description                                                                                                                                                                                                                       | Module / Component                                          | Status (Done/In Progress /Pending) | Marks Contribution |
|-------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------|--------------------|
| 4     | <b>ML Model Training &amp; Comparison</b>          | Train and evaluate 4 classification algorithms: Logistic Regression (56.7%), Decision Tree (68.6%), Random Forest (74.5%), and XGBoost (78.9%). Compare accuracy, precision, recall, and F1-score. Select XGBoost as best model.          | 04_model_building.ipynb Epic 4: Model Building              | Done                               | Core Feature       |
| 5     | <b>Single Applicant Prediction Web Form</b>        | Flask web application with an HTML form accepting 17 applicant inputs (gender, income, age, employment, education etc.). Returns instant Approved/Rejected prediction with a risk score percentage on the same page.                      | app.py + index.html Epic 5: Application Building (/predict) | Done                               | Core Feature       |
| 6     | <b>Batch CSV Screening for Compliance Officers</b> | Upload a CSV file containing multiple applicants and screen them all at once. The system processes every row, runs the XGBoost model on each applicant, and returns a results table with applicant ID, approval status, and risk score %. | app.py + batch.html + batch_results.html (/batch_predict)   | Done                               | Additional Feature |
| 7     | <b>Model Persistence &amp; Loading</b>             | Save trained XGBoost model as credit_approval_model.pkl and feature column order as model_columns.pkl using Joblib. Flask application loads both files at startup for instant, stateless real-time inference without retraining.          | credit_approval_model.pkl model_columns.pkl                 | Done                               | Core Feature       |
| 8     | <b>IBM Watson ML Cloud Deployment Pipeline</b>     | Complete IBM Watson Machine Learning deployment pipeline documented in 05_watson_deployment.ipynb — covers model upload to Watson ML repository, online endpoint creation, scoring API integration with Flask application.                | 05_watson_deployment.ipynb Epic 5: Cloud Deploy             | Done                               | Additional Feature |

### Feature Summary:

| Metric                             | Count / Value |
|------------------------------------|---------------|
| Total Features Planned             | 8             |
| Total Features Implemented         | 8             |
| Core / Must-Have Features          | 6             |
| Additional / Nice-to-Have Features | 2             |
| Features Tested & Verified         | 8             |

### Feature Category Breakdown:

| S. No | Category                  | Features in Category | Example Features                                                                                                          |
|-------|---------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1     | User Interface (UI)       | 3                    | Single prediction form (index.html), Batch upload page (batch.html), Batch results table (batch_results.html)             |
| 2     | Backend / Logic           | 4                    | Feature engineering pipeline, XGBoost inference, model loading (Joblib), Flask route handlers (/predict, /batch_predict)  |
| 3     | Data / Storage            | 3                    | Kaggle CSV datasets (application_record + credit_record), saved model PKL files, watson_model_pipeline.pkl                |
| 4     | API / Integration         | 2                    | IBM Watson ML REST API endpoint, Flask HTTP routes as internal API (/predict POST, /batch_predict POST)                   |
| 5     | Security / Authentication | 1                    | Route-based access separation (/ for single, /batch for compliance), IBM Watson API key authentication for cloud endpoint |